k8s bootcamp - day2 homework

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Labels	
AI summary	Kubernetes CD assignment guides building a Jenkins pipeline that checks out code, validates, builds and pushes a Docker image to Aliyun, then deploys it to a Kubernetes cluster, verifies rollout, and supports rollbacks; it includes project structure, sample app, detailed pipeline script, credential setup, and step-by-step instructions for deploying version v1, updating to v2, and verifying the deployments.

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Deploy version v1

Deploy version v2

Kubernetes Continuous Deployment Assignment with Jenkins

Objective

Build a Jenkins pipeline that automatically:

1. Checks out code from Git.
2. Validates the application.
3. Builds a Docker image.
4. Pushes the image to a container registry.
5. Deploys the image to Kubernetes.
6. Verifies the rollout.
7. Supports rollback.

Project Structure

```
kubernetes-cd-demo/
├── app/
│   ├── index.html
│   └── Dockerfile
├── k8s/
│   ├── namespace.yaml
│   ├── deployment.yaml
│   ├── service.yaml
│   └── ingress.yaml
└── Jenkinsfile
```

Task 1: Prepare the Application

app/index.html

```
<!DOCTYPE html>
<html>
<body>
  <h1>Kubernetes Continuous Deployment Demo</h1>
  <p>Application version: v1</p>
</body>
</html>
```

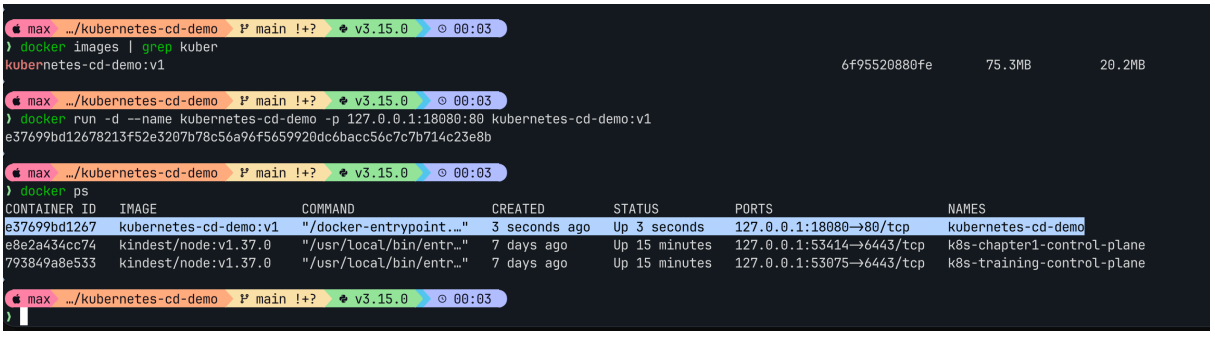
app/Dockerfile

```
FROM nginx:1.25-alpine
COPY index.html /usr/share/nginx/html/index.html
EXPOSE 80
```

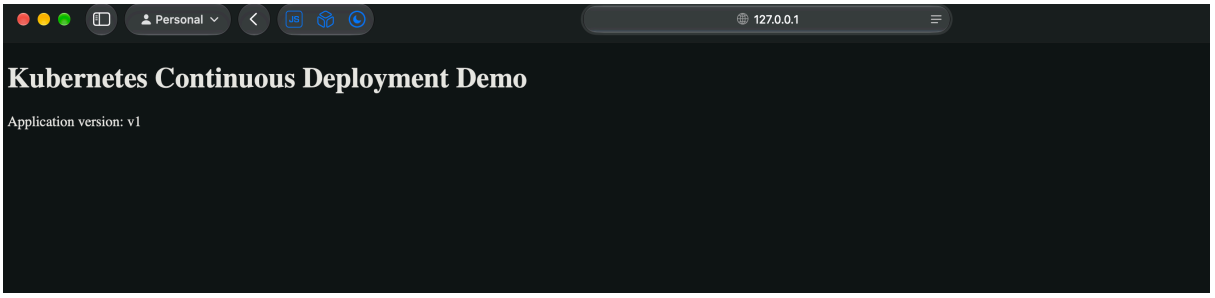
build docker image as below



start the container app



Access the web portal in browser



Task 2: Kubernetes Configuration

Use the Kubernetes YAML files from the previous assignment:

- `namespace.yaml`
- `deployment.yaml`
- `service.yaml`
- `ingress.yaml`

Update the Deployment image to:

```
image: YOUR_REGISTRY/kubernetes-cd-demo:v1
```

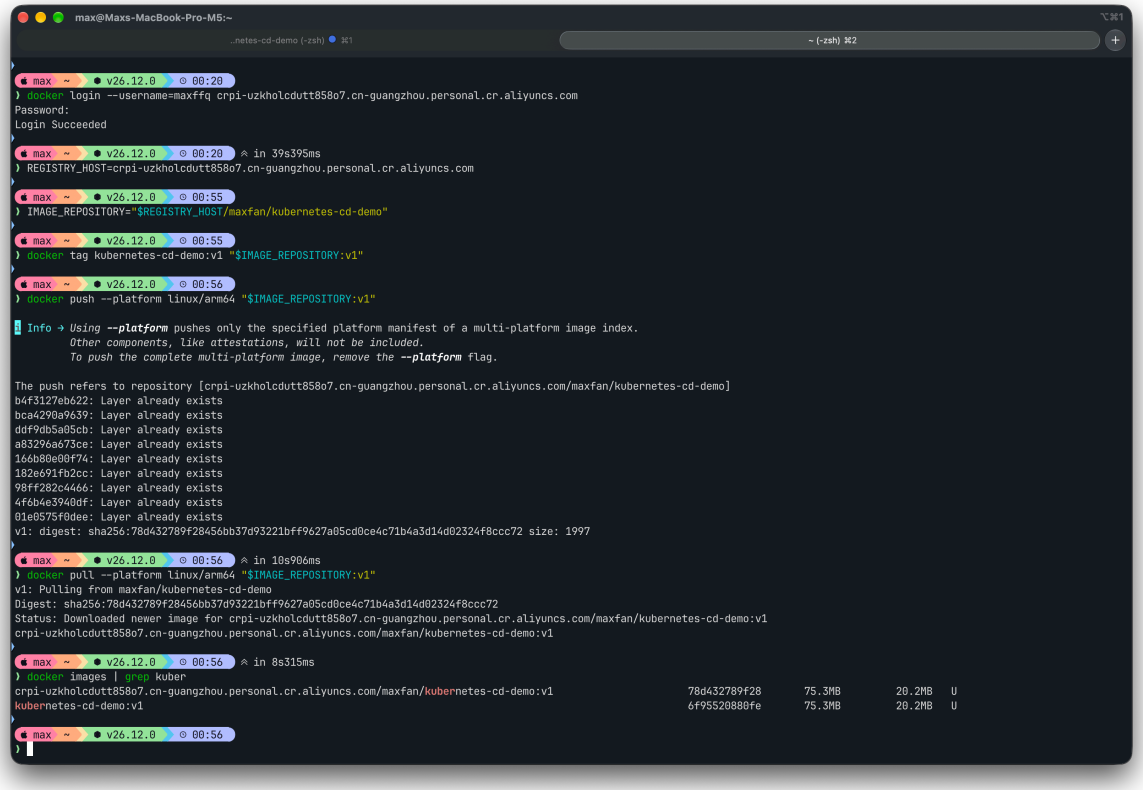
here , i push the docker image to remote repository (aliyun) , then i can define my own YOUR_REGISTRY

```
docker login --username=maxffq crpi-uzkholcdutt858o7.cn-guangzhou.personal.cr.aliyuncs.com
REGISTRY_HOST=crpi-uzkholcdutt858o7.cn-guangzhou.personal.cr.aliyuncs.com
IMAGE_REPOSITORY="$REGISTRY_HOST/maxfan/kubernetes-cd-demo"

docker tag kubernetes-cd-demo:v1 "$IMAGE_REPOSITORY:v1"

docker push --platform linux/arm64 "$IMAGE_REPOSITORY:v1"
```

```
docker pull --platform linux/arm64 "$IMAGE_REPOSITORY:v1"
```

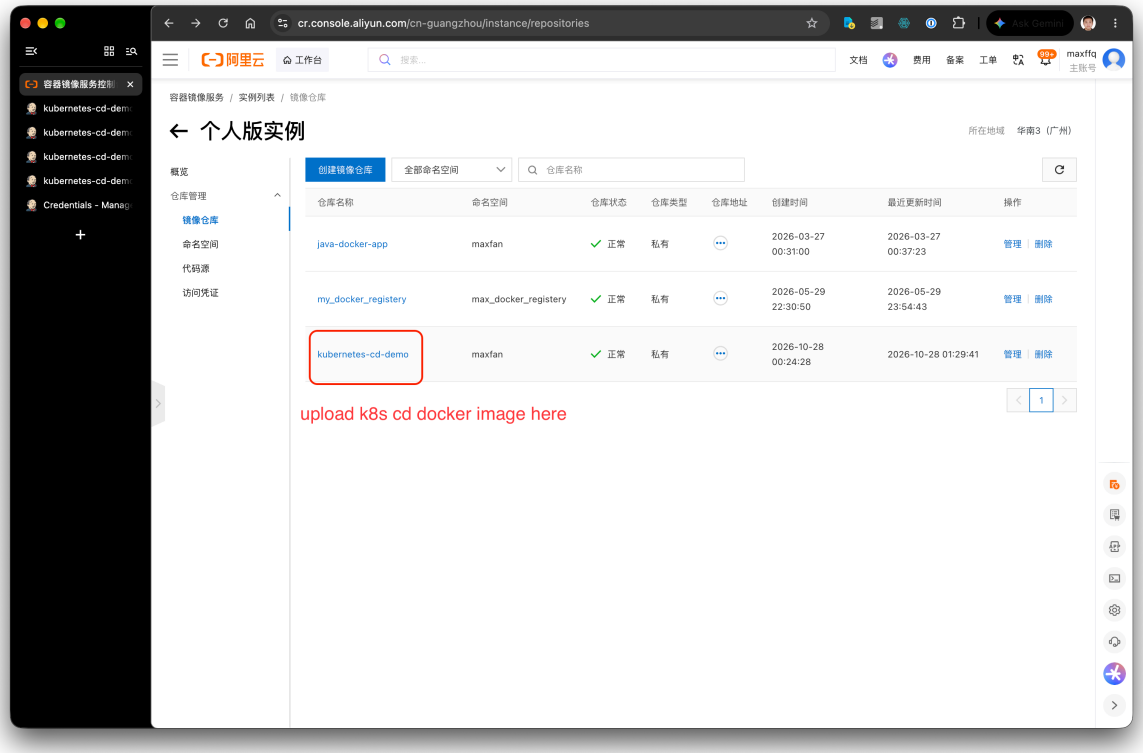


Verify it in aliyun

阿里云登录 - 欢迎登录阿里云，安全稳定的云计算服务平台

欢迎登录阿里云，全球领先的云计算及人工智能科技公司，阿里云为200多个国家和地区的企业、开发者和政府机构提供云计算基础服务及解决方案。阿里云云计算、安全、大数据、人工智能、企业

https://cr.console.aliyun.com/cn-guangzhou/instance/repositories



Task 3: Jenkinsfile

Create a `Jenkinsfile` in the project root:

```
pipeline {
    agent { label 'built-in' }

    environment {
        PATH = "/opt/homebrew/bin:/usr/local/bin:${env.PAT
H}"
        REGISTRY_HOST = 'crpi-uzkholcdutt858o7.cn-guangzho
u.personal.cr.aliyuncs.com'
        IMAGE_REPOSITORY = 'crpi-uzkholcdutt858o7.cn-guangz
hou.personal.cr.aliyuncs.com/maxfan/kubernetes-cd-demo'
        IMAGE_PLATFORM = 'linux/arm64'
        KUBE_CONTEXT = 'kind-k8s-chapter1'
        KUBE_NAMESPACE = 'training'
        DEPLOYMENT_NAME = 'cd-demo-deployment'
        CONTAINER_NAME = 'cd-demo'
    }
```

```
    options {
        timestamps()
        skipDefaultCheckout(true)
        disableConcurrentBuilds()
        timeout(time: 20, unit: 'MINUTES')
    }

    stages {
        stage('Checkout') {
            steps {
                deleteDir()
                sh '''
                    set -eu
                    source_dir=/Users/max/PycharmProjects/P
ythonProject/kubernetes-cd-demo
                    cp -R "$source_dir/app" "$source_dir/k8
s" .
                    cp "$source_dir/Jenkinsfile" .
                    git init -b main
                    git add app k8s Jenkinsfile
                    git -c user.name='Jenkins Local Test' \
                        -c user.email='jenkins-local@exampl
e.invalid' \
                        commit -m 'Verify the pipeline with
out publishing source code' \
                        -m 'Use a local workspace snapshot
to test the deployment stages before SCM publication.' \
                        -m 'Constraint: Do not upload sourc
e code to GitHub
Confidence: high
Scope-risk: narrow
Tested: Pending Jenkins script test'
                '''
                script {
                    env.IMAGE_TAG = sh(
                        script: 'git rev-parse --short=12 H
EAD',
                        returnStdout: true
```

```
                    ).trim()
                }
            }
        }

        stage('Validate') {
            steps {
                sh '''
                    set -eu
                    for file in app/index.html app/Dockerfi
le k8s/namespace.yaml \
                        k8s/deployment.yaml k8s/service.yam
l k8s/ingress.yaml; do
                        test -s "$file"
                    done
                    grep -q 'Application version:' app/inde
x.html
                    git --version
                    docker version
                    docker buildx version
                    kubectl version --client
                '''
            }
        }

        stage('Build and Push Image') {
            steps {
                sh '''
                    set -eu
                    docker buildx build --platform "$IMAGE_
PLATFORM" \
                        --provenance=false --sbom=false --l
oad \
                        -t "$IMAGE_REPOSITORY:$IMAGE_TAG"
./app
                '''
                withCredentials([
                    usernamePassword(
```

```
                        credentialsId: 'container-registry-
credentials',
                        usernameVariable: 'REGISTRY_USERNAM
E',
                        passwordVariable: 'REGISTRY_PASSWOR
D'
                    )
                ]) {
                    sh '''
                        set +x
                        set -eu
                        # Preserve the Docker Desktop endpo
int before isolating login data.
                        export DOCKER_HOST="${DOCKER_HOST:
-$(docker context inspect --format '{{.Endpoints.docker.Hos
t}}')}"
                        unset DOCKER_CONTEXT
                        DOCKER_CONFIG=$(mktemp -d)
                        export DOCKER_CONFIG
                        trap 'docker logout "$REGISTRY_HOS
T" >/dev/null 2>&1 || true; rm -rf "$DOCKER_CONFIG"' EXIT
                        printf '%s' "$REGISTRY_PASSWORD" |
docker login "$REGISTRY_HOST" \
                            --username "$REGISTRY_USERNAME"
--password-stdin
                        docker push --platform "$IMAGE_PLAT
FORM" "$IMAGE_REPOSITORY:$IMAGE_TAG"
                    '''
                }
            }
        }

        stage('Deploy to Kubernetes') {
            steps {
                withCredentials([
                    file(credentialsId: 'kubeconfig-trainin
g', variable: 'KUBECONFIG_FILE')
                ]) {
```

```
                    sh '''
                        set +x
                        set -eu
                        export KUBECONFIG="$KUBECONFIG_FIL
E"
                        kubectl --context="$KUBE_CONTEXT" a
pply -f k8s/namespace.yaml
                        kubectl --context="$KUBE_CONTEXT" -
n "$KUBE_NAMESPACE" \
                            get secret cd-demo-acr-pull -o
name

                        deployment_file=$(mktemp)
                        trap 'rm -f "$deployment_file"' EXI
T
                        kubectl set image --local -f k8s/de
ployment.yaml \
                            "$CONTAINER_NAME=$IMAGE_REPOSIT
ORY:$IMAGE_TAG" \
                            -o yaml > "$deployment_file"
                        kubectl --context="$KUBE_CONTEXT" a
pply \
                            -f k8s/service.yaml -f k8s/ingr
ess.yaml -f "$deployment_file"
                        kubectl --context="$KUBE_CONTEXT" -
n "$KUBE_NAMESPACE" rollout status \
                            deployment/"$DEPLOYMENT_NAME" -
-timeout=180s
                    '''
                }
            }
        }

        stage('Verify') {
            steps {
                withCredentials([
                    file(credentialsId: 'kubeconfig-trainin
g', variable: 'KUBECONFIG_FILE')
```

```
                ]) {
                    sh '''
                        set +x
                        set -eu
                        export KUBECONFIG="$KUBECONFIG_FIL
E"
                        actual_image=$(kubectl --context
="$KUBE_CONTEXT" -n "$KUBE_NAMESPACE" \
                            get deployment "$DEPLOYMENT_NAM
E" \
                            -o "jsonpath={.spec.template.sp
ec.containers[?(@.name=='$CONTAINER_NAME')].image}")
                        test "$actual_image" = "$IMAGE_REPO
SITORY:$IMAGE_TAG"
                        kubectl --context="$KUBE_CONTEXT" -
n "$KUBE_NAMESPACE" \
                            get deployments,pods,services,i
ngresses -l app=cd-demo -o wide
                        kubectl --context="$KUBE_CONTEXT" -
n "$KUBE_NAMESPACE" \
                            get endpointslices -l kubernete
s.io/service-name=cd-demo-service
                        kubectl --context="$KUBE_CONTEXT" -
n "$KUBE_NAMESPACE" \
                            rollout history deployment/"$DE
PLOYMENT_NAME"
                    '''
                }
            }
        }
    }

    post {
        success {
            echo "Deployment completed successfully: ${env.
IMAGE_REPOSITORY}:${env.IMAGE_TAG}"
        }
        failure {
```

```
            echo 'Deployment failed. Check the Jenkins cons
ole and Kubernetes events before retrying.'
        }
    }
}
```

Task 4: Configure Jenkins

Create a Jenkins Pipeline job using **Pipeline script from SCM**.

Configure these credentials:

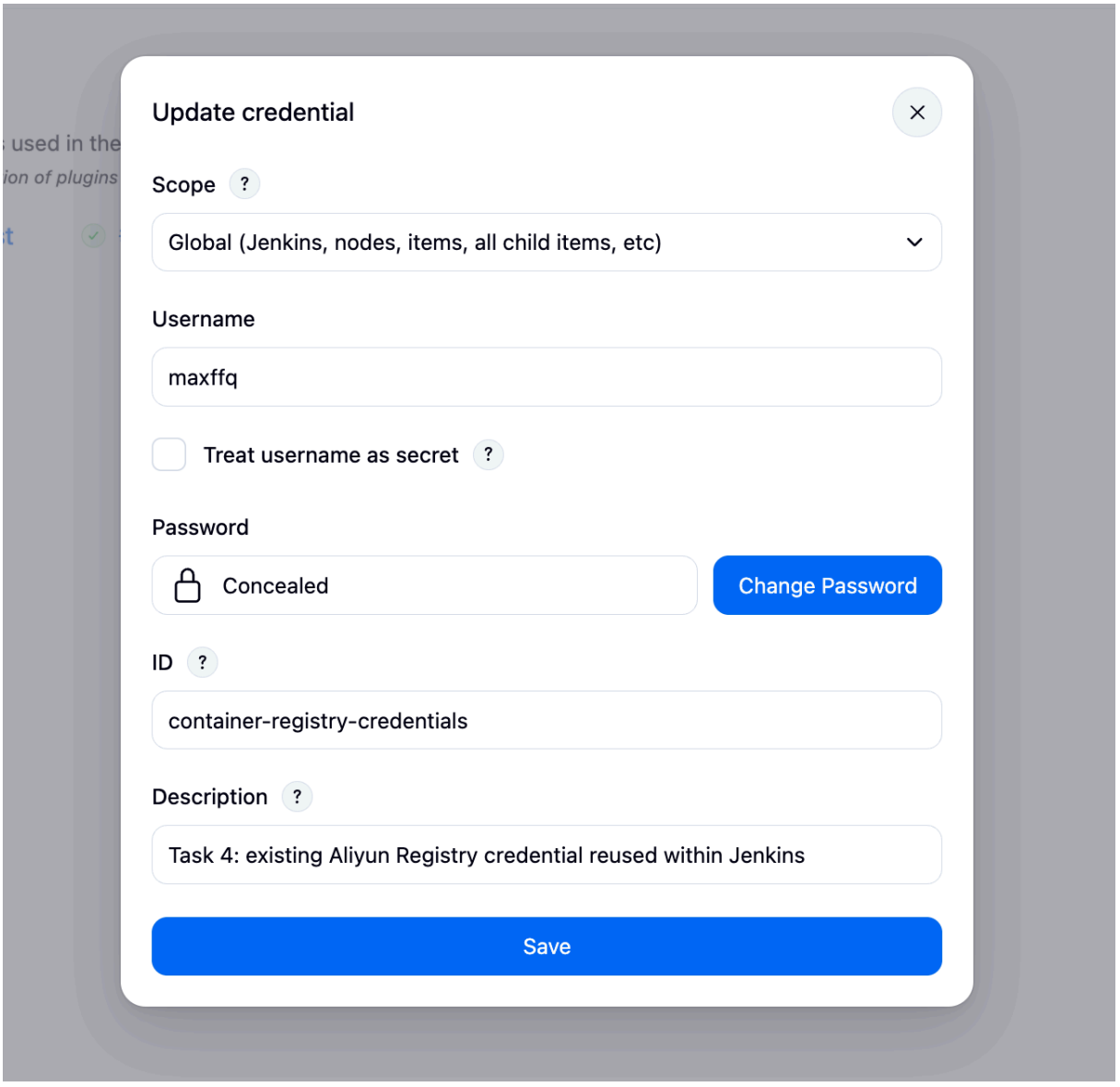
Credential ID	Type	Purpose
`container-registry-credentials`	Username and password	Log in to the container registry
`kubeconfig-training`	Secret file	Access the Kubernetes cluster

Do not commit passwords, access tokens or kubeconfig files to Git.

create credential container-registry-credentials

This is the credential used in aliyun.com

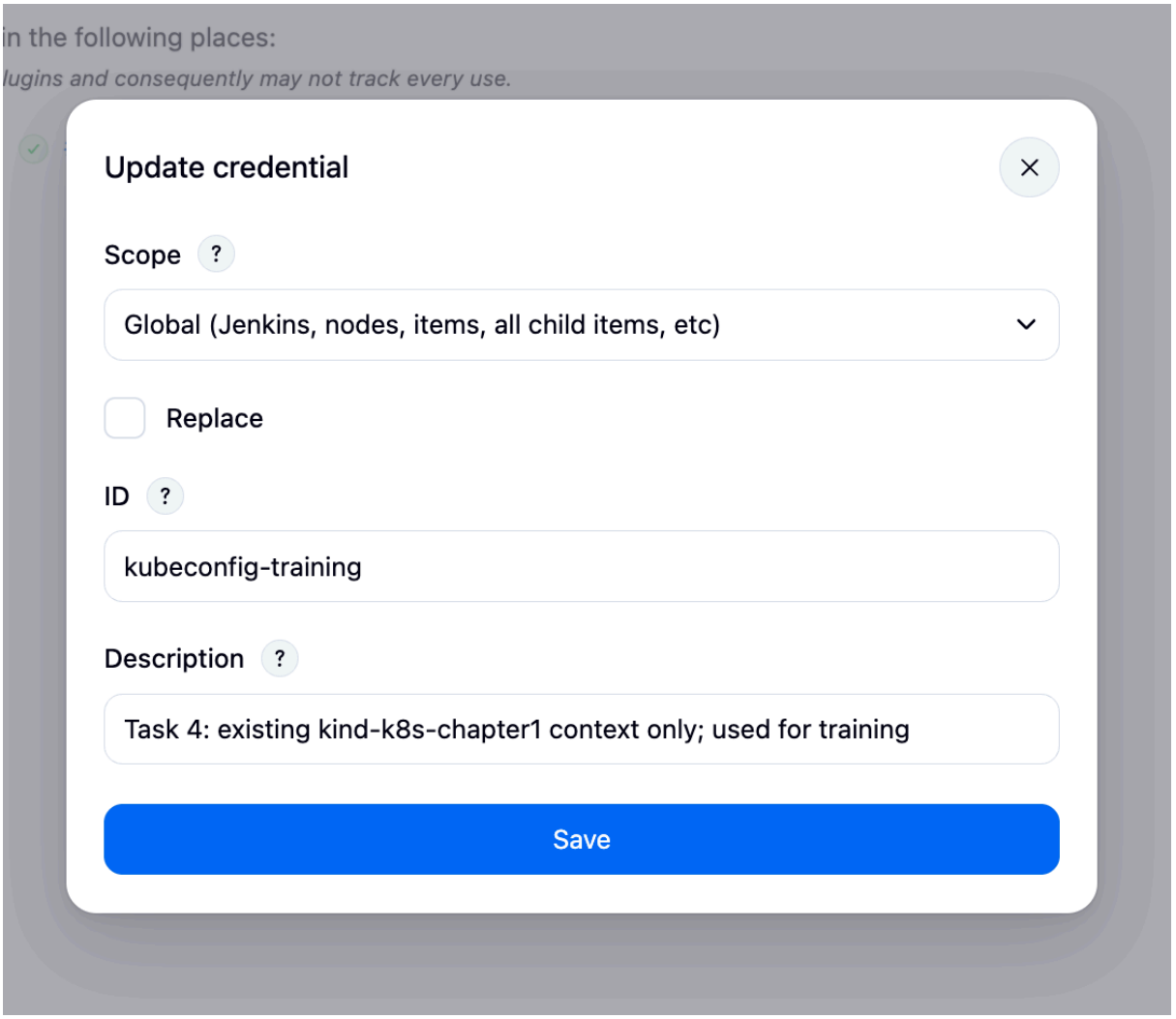
we use the credential to push or pull docker image from aliyun



Create credential kubeconfig-training

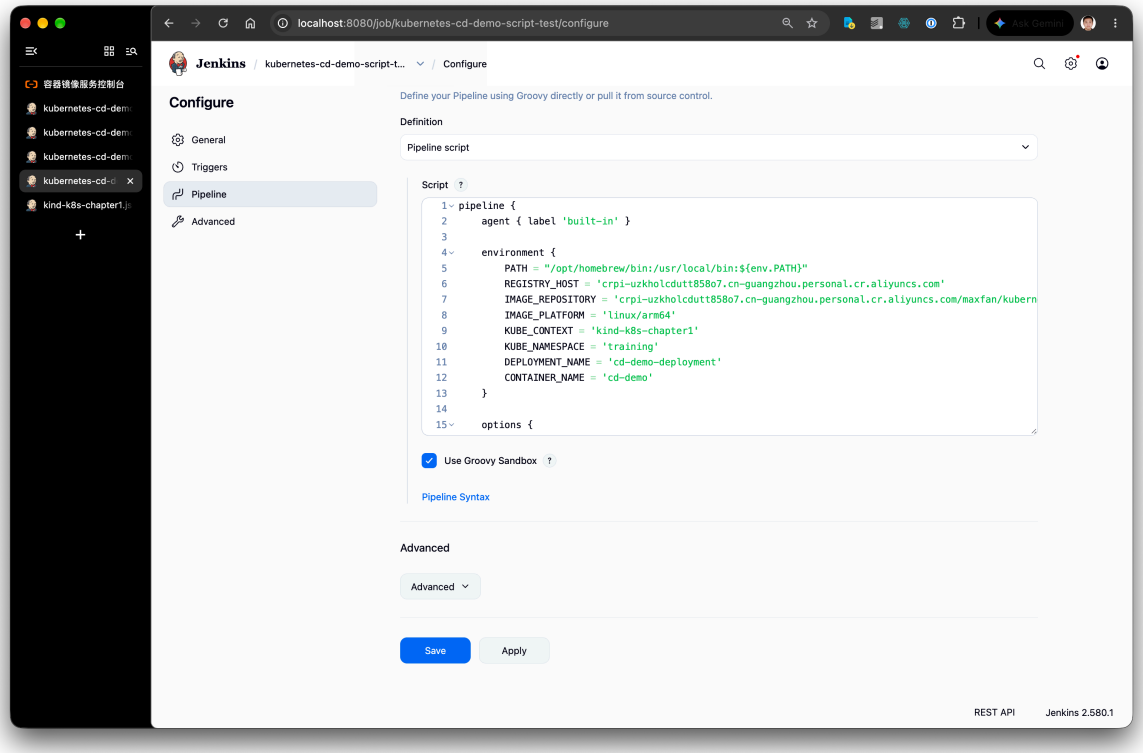
We can run following command to output the result to a file , then upload the file to the jenkins to create the credential

```
kubectl config view --raw
```

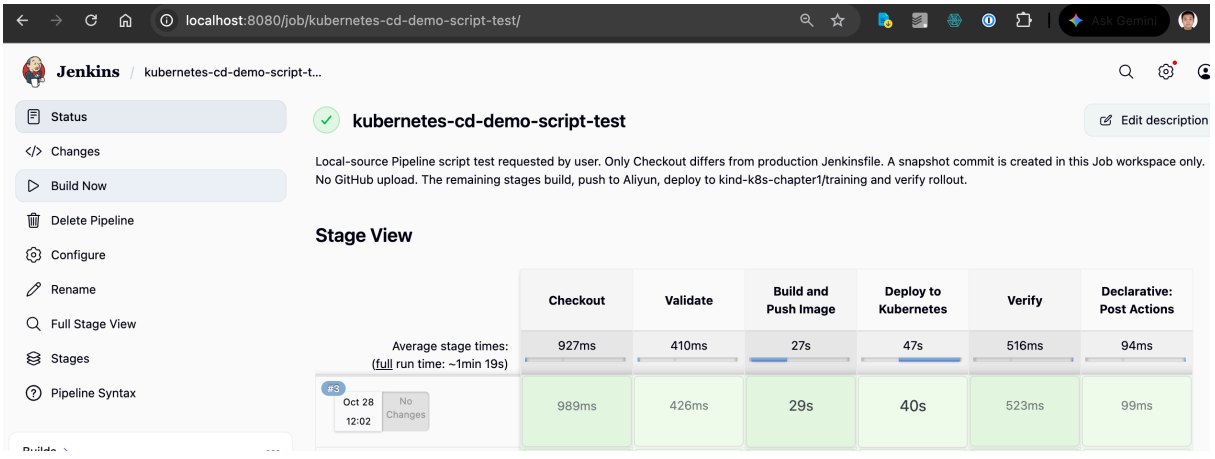


Create one jenkins pipeline to test it

Create a jenkins pipeline, copy the Jenkinsfile content into it



Trigger the jenkins build



find logs

```
Started by user max
[Pipeline] Start of Pipeline
```

```
[Pipeline] node
Running on Jenkins in /Users/max/.jenkins/workspace/kuberne
tes-cd-demo-script-test
[Pipeline] {
[Pipeline] withEnv
[Pipeline] {
[Pipeline] timestamps
[Pipeline] {
[Pipeline] timeout
12:02:53  Timeout set to expire in 20 min
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Checkout)
[Pipeline] deleteDir
[Pipeline] sh
12:02:53  + set -eu
12:02:53  + source_dir=/Users/max/PycharmProjects/PythonPro
ject/kubernetes-cd-demo
12:02:53  + cp -R /Users/max/PycharmProjects/PythonProject/
kubernetes-cd-demo/app /Users/max/PycharmProjects/PythonPro
ject/kubernetes-cd-demo/k8s .
12:02:53  + cp /Users/max/PycharmProjects/PythonProject/kub
ernetes-cd-demo/Jenkinsfile .
12:02:53  + git init -b main
12:02:53  Initialized empty Git repository in /Users/max/.j
enkins/workspace/kubernetes-cd-demo-script-test/.git/
12:02:53  + git add app k8s Jenkinsfile
12:02:53  + git -c 'user.name=Jenkins Local Test' -c user.e
mail=jenkins-local@example.invalid commit -m 'Verify the pi
peline without publishing source code' -m 'Use a local work
space snapshot to test the deployment stages before SCM pub
lication.' -m 'Constraint: Do not upload source code to Git
Hub
12:02:53  Confidence: high
12:02:53  Scope-risk: narrow
12:02:53  Tested: Pending Jenkins script test'
12:02:53  [main (root-commit) f7a5d2d] Verify the pipeline
without publishing source code
```

```
12:02:53   7 files changed, 234 insertions(+)
12:02:53   create mode 100644 Jenkinsfile
12:02:53   create mode 100644 app/Dockerfile
12:02:53   create mode 100644 app/index.html
12:02:53   create mode 100644 k8s/deployment.yaml
12:02:53   create mode 100644 k8s/ingress.yaml
12:02:53   create mode 100644 k8s/namespace.yaml
12:02:53   create mode 100644 k8s/service.yaml
[Pipeline] script
[Pipeline] {
[Pipeline] sh
12:02:54  + git rev-parse --short=12 HEAD
[Pipeline] }
[Pipeline] // script
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Validate)
[Pipeline] sh
12:02:54  + set -eu
12:02:54  + for file in app/index.html app/Dockerfile k8s/n
amespace.yaml k8s/deployment.yaml k8s/service.yaml k8s/ingr
ess.yaml
12:02:54  + test -s app/index.html
12:02:54  + for file in app/index.html app/Dockerfile k8s/n
amespace.yaml k8s/deployment.yaml k8s/service.yaml k8s/ingr
ess.yaml
12:02:54  + test -s app/Dockerfile
12:02:54  + for file in app/index.html app/Dockerfile k8s/n
amespace.yaml k8s/deployment.yaml k8s/service.yaml k8s/ingr
ess.yaml
12:02:54  + test -s k8s/namespace.yaml
12:02:54  + for file in app/index.html app/Dockerfile k8s/n
amespace.yaml k8s/deployment.yaml k8s/service.yaml k8s/ingr
ess.yaml
12:02:54  + test -s k8s/deployment.yaml
12:02:54  + for file in app/index.html app/Dockerfile k8s/n
amespace.yaml k8s/deployment.yaml k8s/service.yaml k8s/ingr
```

```
ess.yaml
12:02:54  + test -s k8s/service.yaml
12:02:54  + for file in app/index.html app/Dockerfile k8s/n
amespace.yaml k8s/deployment.yaml k8s/service.yaml k8s/ingr
ess.yaml
12:02:54  + test -s k8s/ingress.yaml
12:02:54  + grep -q 'Application version:' app/index.html
12:02:54  + git --version
12:02:54  git version 2.54.0 (Apple Git-157)
12:02:54  + docker version
12:02:54  Client:
12:02:54   Version:           29.9.2
12:02:54   API version:       1.56
12:02:54   Go version:        go1.26.10
12:02:54   Git commit:        a82e528
12:02:54   Built:             Tue Oct 20 18:43:59 2026
12:02:54   OS/Arch:           darwin/arm64
12:02:54   Context:           desktop-linux
12:02:54
12:02:54  Server: Docker Desktop 4.96.1 (244132)
12:02:54   Engine:
12:02:54    Version:          29.9.2
12:02:54    API version:      1.56 (minimum version 1.40)
12:02:54    Go version:       go1.26.10
12:02:54    Git commit:       0367cbe
12:02:54    Built:            Tue Oct 20 18:44:14 2026
12:02:54    OS/Arch:          linux/arm64
12:02:54    Experimental:     false
12:02:54   containerd:
12:02:54    Version:          v2.4.2
12:02:54    GitCommit:        b7a25261e029d68978bbf290df0a6
2fc819b3c51
12:02:54   runc:
12:02:54    Version:          1.5.2
12:02:54    GitCommit:        v1.5.2-0-g29dd3dc2
12:02:54   docker-init:
12:02:54    Version:          0.19.0
12:02:54    GitCommit:        de40ad0
```

```
12:02:54  + docker buildx version
12:02:54  github.com/docker/buildx v0.38.1 0dbe87f36ed472fe
36b6796c3f70b4210a731fc5
12:02:54  + kubectl version --client
12:02:54  Client Version: v1.36.1
12:02:54  Kustomize Version: v5.8.1
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build and Push Image)
[Pipeline] sh
12:02:55  + set -eu
12:02:55  + docker buildx build --platform linux/arm64 --pr
ovenance=false --sbom=false --load -t crpi-uzkholcdutt858o
7.cn-guangzhou.personal.cr.aliyuncs.com/maxfan/kubernetes-c
d-demo:f7a5d2d6dd33 ./app
12:02:55  #0 building with "desktop-linux" instance using d
ocker driver
12:02:55
12:02:55  #1 [internal] load build definition from Dockerfi
le
12:02:55  #1 transferring dockerfile: 119B done
12:02:55  #1 DONE 0.0s
12:02:55
12:02:55  #2 [internal] load metadata for docker.io/librar
y/nginx:1.25-alpine
12:02:57  #2 ...
12:02:57
12:02:57  #3 [auth] library/nginx:pull token for registry-
1.docker.io
12:02:57  #3 DONE 0.0s
12:02:57
12:02:57  #2 [internal] load metadata for docker.io/librar
y/nginx:1.25-alpine
12:03:00  #2 DONE 4.7s
12:03:00
12:03:00  #4 [internal] load .dockerignore
12:03:00  #4 transferring context: 2B done
```

```
12:03:00  #4 DONE 0.0s
12:03:00
12:03:00  #5 [internal] load build context
12:03:00  #5 transferring context: 167B done
12:03:00  #5 DONE 0.0s
12:03:00
12:03:00  #6 [1/2] FROM docker.io/library/nginx:1.25-alpine
@sha256:516475cc129da42866742567714ddc681e5eed7b9ee0b9e9c01
5e464b4221a00
12:03:00  #6 resolve docker.io/library/nginx:1.25-alpine@sh
a256:516475cc129da42866742567714ddc681e5eed7b9ee0b9e9c015e4
64b4221a00 done
12:03:00  #6 DONE 0.0s
12:03:00
12:03:00  #7 [2/2] COPY index.html /usr/share/nginx/html/in
dex.html
12:03:00  #7 CACHED
12:03:00
12:03:00  #8 exporting to image
12:03:00  #8 exporting layers done
12:03:00  #8 exporting manifest sha256:78d432789f28456bb37d
93221bff9627a05cd0ce4c71b4a3d14d02324f8ccc72 done
12:03:00  #8 exporting config sha256:c10489f14680945f6b8252
92835e03d13006e6f4a2615f9ee2181b9402e5da9d done
12:03:00  #8 naming to crpi-uzkholcdutt858o7.cn-guangzhou.p
ersonal.cr.aliyuncs.com/maxfan/kubernetes-cd-demo:f7a5d2d6d
d33 done
12:03:00  #8 unpacking to crpi-uzkholcdutt858o7.cn-guangzho
u.personal.cr.aliyuncs.com/maxfan/kubernetes-cd-demo:f7a5d2
d6dd33 done
12:03:00  #8 DONE 0.0s
[Pipeline] withCredentials
12:03:00  Masking supported pattern matches of $REGISTRY_PA
SSWORD
[Pipeline] {
[Pipeline] sh
12:03:00  + set +x
12:03:16  Login Succeeded
```

```
12:03:16
12:03:16   Info -> Using --platform pushes only the specifi
ed platform manifest of a multi-platform image index.
12:03:16           Other components, like attestations, wil
l not be included.
12:03:16           To push the complete multi-platform imag
e, remove the --platform flag.
12:03:16
12:03:16  The push refers to repository [crpi-uzkholcdutt85
8o7.cn-guangzhou.personal.cr.aliyuncs.com/maxfan/kubernetes
-cd-demo]
12:03:16  b4f3127eb622: Waiting
12:03:16  bca4290a9639: Waiting
12:03:16  ddf9db5a05cb: Waiting
12:03:16  a83296a673ce: Waiting
12:03:16  b4f3127eb622: Waiting
12:03:16  bca4290a9639: Waiting
12:03:16  ddf9db5a05cb: Waiting
12:03:16  a83296a673ce: Waiting
12:03:16  b4f3127eb622: Waiting
12:03:16  bca4290a9639: Waiting
12:03:16  ddf9db5a05cb: Waiting
12:03:16  a83296a673ce: Waiting
12:03:16  a83296a673ce: Waiting
12:03:16  b4f3127eb622: Waiting
12:03:16  bca4290a9639: Waiting
12:03:16  ddf9db5a05cb: Waiting
12:03:16  b4f3127eb622: Waiting
12:03:16  bca4290a9639: Waiting
12:03:16  ddf9db5a05cb: Waiting
12:03:16  a83296a673ce: Waiting
12:03:16  b4f3127eb622: Waiting
12:03:16  bca4290a9639: Waiting
12:03:16  ddf9db5a05cb: Waiting
12:03:16  a83296a673ce: Waiting
12:03:16  b4f3127eb622: Waiting
12:03:16  bca4290a9639: Waiting
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12:03:21  ddf9db5a05cb: Waiting
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12:03:21  a83296a673ce: Waiting
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12:03:21  ddf9db5a05cb: Waiting
12:03:21  bca4290a9639: Waiting
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12:03:21  ddf9db5a05cb: Waiting
12:03:21  a83296a673ce: Layer already exists
12:03:21  166b80e00f74: Waiting
12:03:21  182e691fb2cc: Waiting
12:03:21  b4f3127eb622: Layer already exists
12:03:21  166b80e00f74: Waiting
12:03:21  182e691fb2cc: Waiting
12:03:21  bca4290a9639: Waiting
12:03:21  ddf9db5a05cb: Waiting
12:03:21  182e691fb2cc: Waiting
12:03:21  bca4290a9639: Waiting
12:03:21  ddf9db5a05cb: Waiting
12:03:21  166b80e00f74: Waiting
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12:03:21  ddf9db5a05cb: Waiting
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12:03:21  ddf9db5a05cb: Waiting
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12:03:21  182e691fb2cc: Waiting
12:03:21  bca4290a9639: Waiting
12:03:21  ddf9db5a05cb: Layer already exists
12:03:21  4f6b4e3940df: Waiting
12:03:21  166b80e00f74: Waiting
12:03:21  182e691fb2cc: Waiting
12:03:21  98ff282c4466: Waiting
12:03:21  bca4290a9639: Layer already exists
12:03:22  4f6b4e3940df: Waiting
12:03:22  166b80e00f74: Waiting
12:03:22  182e691fb2cc: Waiting
12:03:22  98ff282c4466: Waiting
12:03:22  4f6b4e3940df: Waiting
12:03:22  166b80e00f74: Waiting
12:03:22  182e691fb2cc: Waiting
12:03:22  98ff282c4466: Waiting
12:03:22  4f6b4e3940df: Waiting
12:03:22  166b80e00f74: Waiting
12:03:22  182e691fb2cc: Waiting
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12:03:22  98ff282c4466: Waiting
12:03:22  4f6b4e3940df: Waiting
12:03:22  166b80e00f74: Waiting
12:03:22  182e691fb2cc: Waiting
12:03:22  98ff282c4466: Waiting
12:03:22  166b80e00f74: Waiting
12:03:22  182e691fb2cc: Waiting
12:03:22  98ff282c4466: Waiting
12:03:22  4f6b4e3940df: Waiting
12:03:22  166b80e00f74: Layer already exists
12:03:22  182e691fb2cc: Waiting
12:03:22  98ff282c4466: Waiting
12:03:22  01e0575f0dee: Waiting
12:03:22  4f6b4e3940df: Waiting
12:03:22  182e691fb2cc: Waiting
12:03:22  98ff282c4466: Waiting
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12:03:22  4f6b4e3940df: Waiting
12:03:22  182e691fb2cc: Waiting
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12:03:22  4f6b4e3940df: Waiting
12:03:22  182e691fb2cc: Waiting
12:03:22  4f6b4e3940df: Waiting
12:03:22  182e691fb2cc: Layer already exists
12:03:22  98ff282c4466: Layer already exists
12:03:22  01e0575f0dee: Waiting
12:03:22  01e0575f0dee: Waiting
12:03:22  4f6b4e3940df: Waiting
12:03:23  4f6b4e3940df: Waiting
12:03:23  01e0575f0dee: Waiting
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12:03:23  4f6b4e3940df: Waiting
12:03:23  01e0575f0dee: Waiting
12:03:23  01e0575f0dee: Waiting
12:03:23  4f6b4e3940df: Waiting
12:03:23  01e0575f0dee: Layer already exists
12:03:23  4f6b4e3940df: Layer already exists
12:03:24  f7a5d2d6dd33: digest: sha256:78d432789f28456bb37d
93221bff9627a05cd0ce4c71b4a3d14d02324f8ccc72 size: 1997
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy to Kubernetes)
[Pipeline] withCredentials
12:03:24  Masking supported pattern matches of $KUBECONFIG_
FILE
[Pipeline] {
[Pipeline] sh
12:03:25  + set +x
12:03:25  namespace/training unchanged
12:03:25  secret/cd-demo-acr-pull
12:03:25  service/cd-demo-service unchanged
12:03:25  ingress.networking.k8s.io/cd-demo-ingress unchang
ed
12:03:25  deployment.apps/cd-demo-deployment configured
12:03:25  Waiting for deployment "cd-demo-deployment" rollo
ut to finish: 1 out of 3 new replicas have been updated...
12:03:44  Waiting for deployment "cd-demo-deployment" rollo
ut to finish: 1 out of 3 new replicas have been updated...
12:03:44  Waiting for deployment "cd-demo-deployment" rollo
ut to finish: 1 out of 3 new replicas have been updated...
12:03:44  Waiting for deployment "cd-demo-deployment" rollo
ut to finish: 2 out of 3 new replicas have been updated...
12:03:44  Waiting for deployment "cd-demo-deployment" rollo
ut to finish: 2 out of 3 new replicas have been updated...
12:03:54  Waiting for deployment "cd-demo-deployment" rollo
ut to finish: 2 out of 3 new replicas have been updated...
```

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12:03:54  Waiting for deployment "cd-demo-deployment" rollo
ut to finish: 2 out of 3 new replicas have been updated...
12:03:54  Waiting for deployment "cd-demo-deployment" rollo
ut to finish: 1 old replicas are pending termination...
12:03:54  Waiting for deployment "cd-demo-deployment" rollo
ut to finish: 1 old replicas are pending termination...
12:04:05  Waiting for deployment "cd-demo-deployment" rollo
ut to finish: 1 old replicas are pending termination...
12:04:05  deployment "cd-demo-deployment" successfully roll
ed out
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Verify)
[Pipeline] withCredentials
12:04:05  Masking supported pattern matches of $KUBECONFIG_
FILE
[Pipeline] {
[Pipeline] sh
12:04:06  + set +x
12:04:06  NAME                                    READY   UP-T
O-DATE   AVAILABLE   AGE   CONTAINERS   IMAGES
SELECTOR
12:04:06  deployment.apps/cd-demo-deployment   3/3     3
3           10h   cd-demo      crpi-uzkholcdutt858o7.cn-gua
ngzhou.personal.cr.aliyuncs.com/maxfan/kubernetes-cd-demo:f
7a5d2d6dd33   app=cd-demo
12:04:06
12:04:06  NAME                                          READY
STATUS      RESTARTS   AGE   IP            NODE
NOMINATED NODE   READINESS GATES
12:04:06  pod/cd-demo-deployment-7665ccbbfb-7rcbn   0/1
Completed   0          10h   10.244.0.18   k8s-chapter1-con
trol-plane   <none>           <none>
12:04:06  pod/cd-demo-deployment-84f975ffcd-2jhpd   1/1
Running     0          40s   10.244.0.20   k8s-chapter1-con
```

```
trol-plane   <none>           <none>
12:04:06  pod/cd-demo-deployment-84f975ffcd-9dpzr   1/1
Running     0          22s   10.244.0.21   k8s-chapter1-con
trol-plane   <none>           <none>
12:04:06  pod/cd-demo-deployment-84f975ffcd-fm8sw   1/1
Running     0          11s   10.244.0.22   k8s-chapter1-con
trol-plane   <none>           <none>
12:04:06
12:04:06  NAME                       TYPE        CLUSTER-IP
EXTERNAL-IP   PORT(S)   AGE   SELECTOR
12:04:06  service/cd-demo-service   ClusterIP   10.96.255.9
<none>        80/TCP    10h   app=cd-demo
12:04:06
12:04:06  NAME                                          CLASS
HOSTS           ADDRESS       PORTS   AGE
12:04:06  ingress.networking.k8s.io/cd-demo-ingress   nginx
cd-demo.local   10.96.59.62   80      10h
12:04:06  NAME                    ADDRESSTYPE   PORTS   END
POINTS                            AGE
12:04:06  cd-demo-service-2kzgs   IPv4          80      10.
244.0.20,10.244.0.21,10.244.0.22   10h
12:04:06  deployment.apps/cd-demo-deployment
12:04:06  REVISION  CHANGE-CAUSE
12:04:06  1         <none>
12:04:06  2         <none>
12:04:06  3         <none>
12:04:06
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
12:04:06  Deployment completed successfully: crpi-uzkholcdu
tt858o7.cn-guangzhou.personal.cr.aliyuncs.com/maxfan/kubern
etes-cd-demo:f7a5d2d6dd33
[Pipeline] }
```

```
[Pipeline] // stage
[Pipeline] }
[Pipeline] // timeout
[Pipeline] }
[Pipeline] // timestamps
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

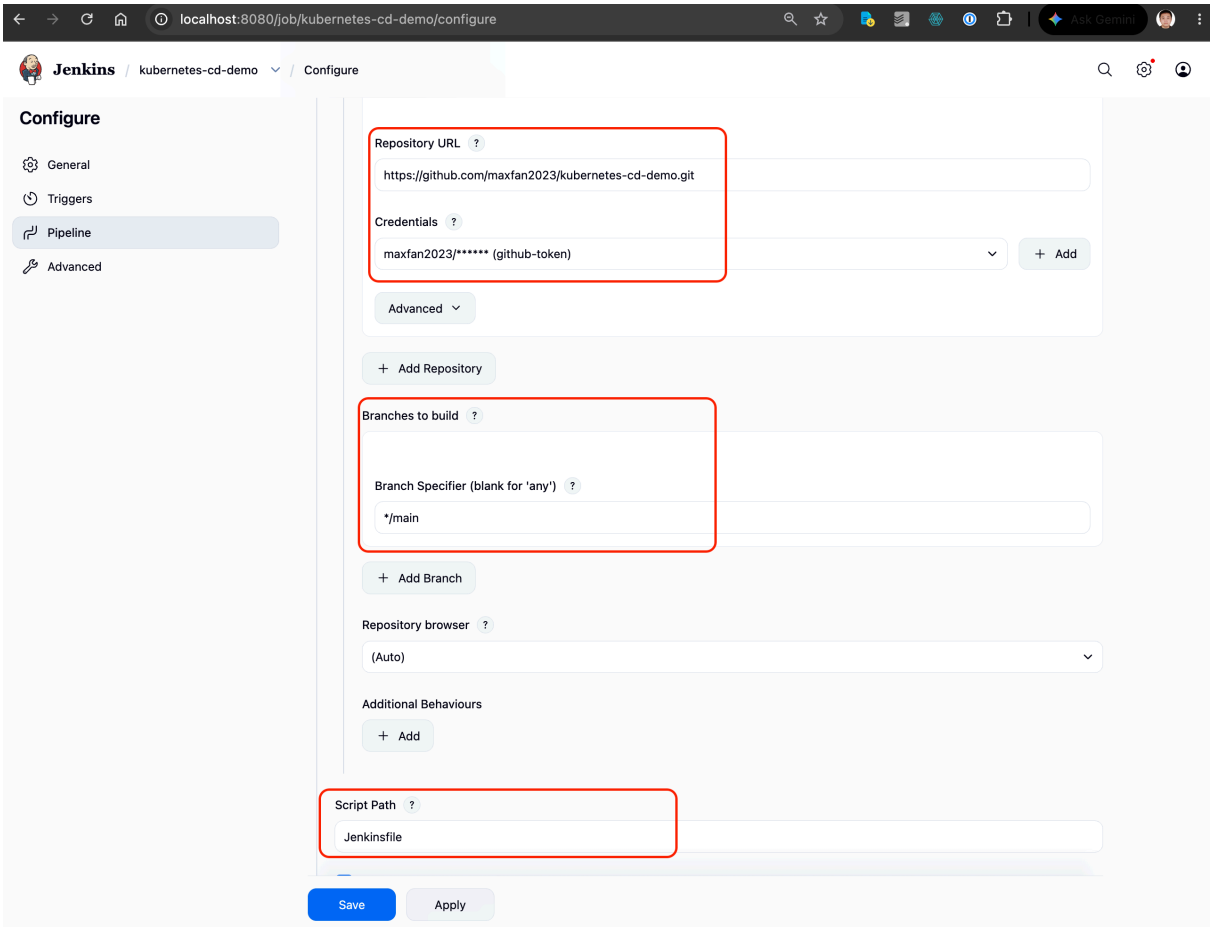
Task 5: Deploy and Update the Application

Deploy version v1

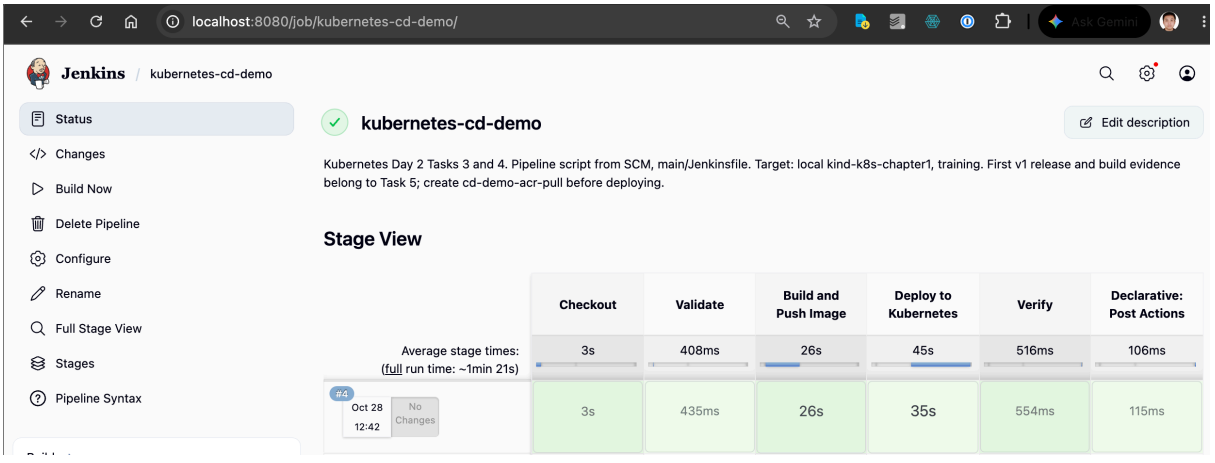
commit the change to github repo https://github.com/maxfan2023/kubernetes-cd-demo

```
git add .
git commit -m "Deploy version v1"
git push origin main
```

create a jenkins file



trigger jenkins build



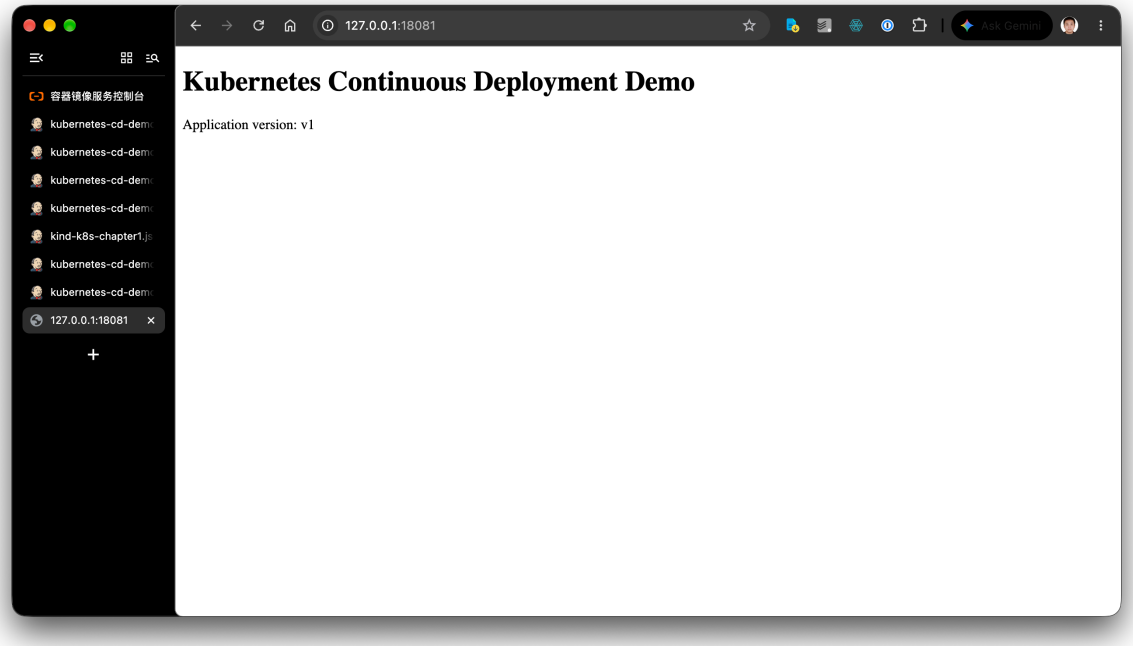
check deployment status

```
❯ kubectl --context=kind-k8s-chapter1 -n training rollout status deployment/cd-demo-deployment
deployment "cd-demo-deployment" successfully rolled out

 max  ~  v26.12.0  12:55
❯ kubectl --context=kind-k8s-chapter1 -n training get pods -l app=cd-demo
NAME                                 READY   STATUS    RESTARTS   AGE
cd-demo-deployment-86d87f45f7-6mh7h   1/1     Running   0          13m
cd-demo-deployment-86d87f45f7-dlm5l   1/1     Running   0          12m
cd-demo-deployment-86d87f45f7-fjnhp   1/1     Running   0          12m

 max  ~  v26.12.0  12:55
❯
```

Access it from web browser



Deploy version v2

Update `index.html`:

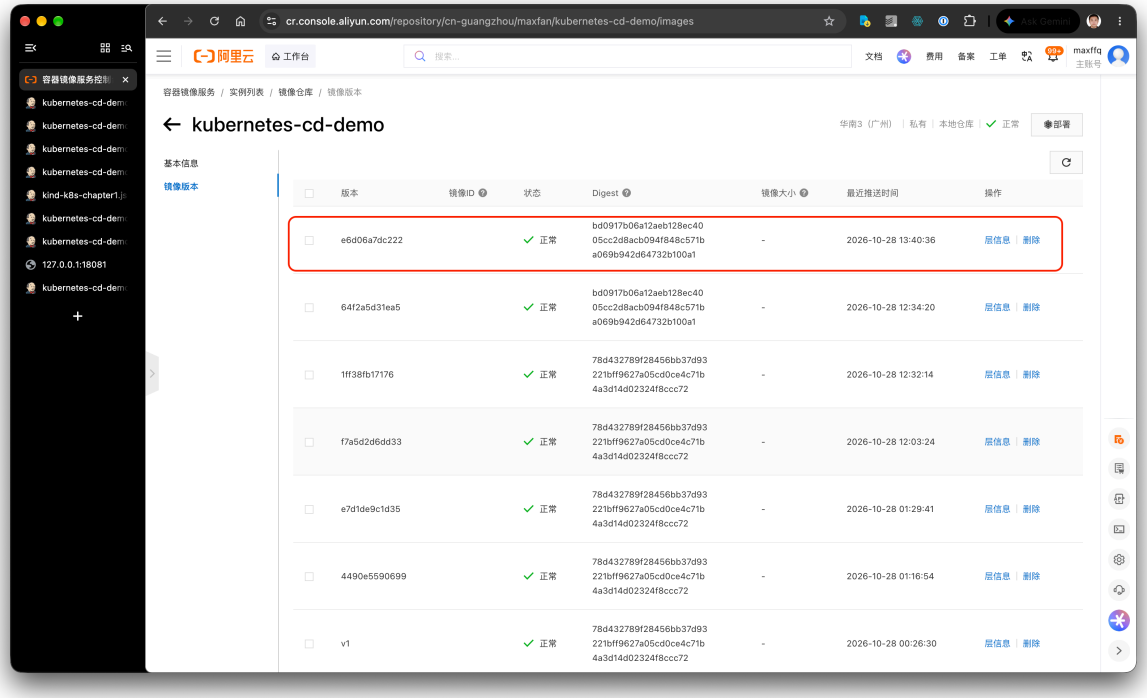
```
<p>Application version: v2</p>
```

Commit and push the change:

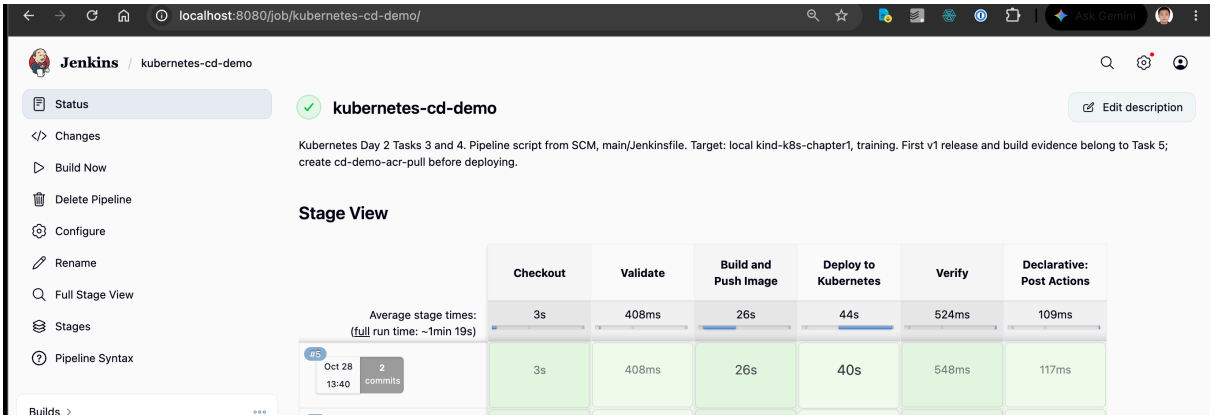
```
git add .
git commit -m "Release version v2"
```

```
git push origin main
```

upload the new docker image to aliyun



trigger jenkins build



access it from web browser

